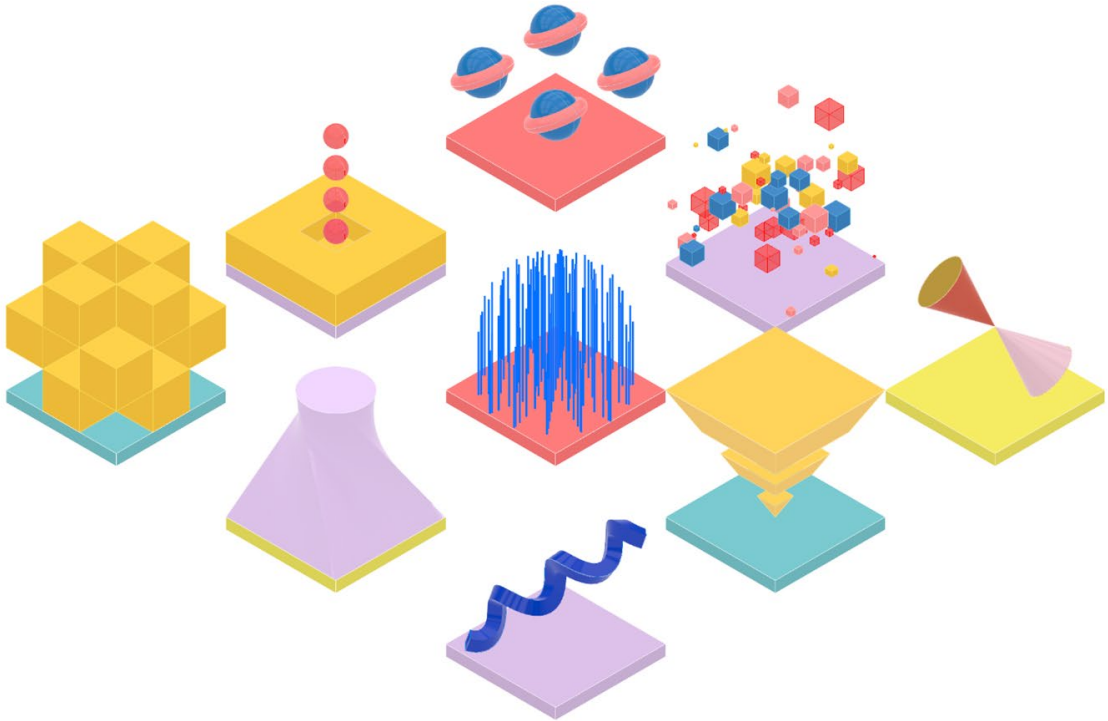
Examples

Lecture 2.1 | 26 October 2026



Let’s go to Rhino/Grasshopper!

- Make sure to update to the latest changes of our GitHub repository by going to **GitHub Desktop** and on top right first clicking first *fetch origin* and then *pull*!
- Make sure you completed the installation steps for Rhino/Grasshopper!
- Open up the file: *cdf1_teaching\exercises\geometry-basics\geometry_basics.ghx*

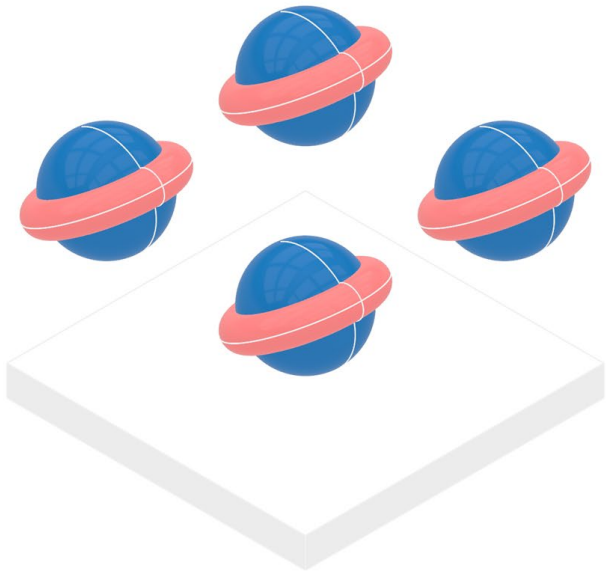




1. Spheres and toruses

Learning objectives

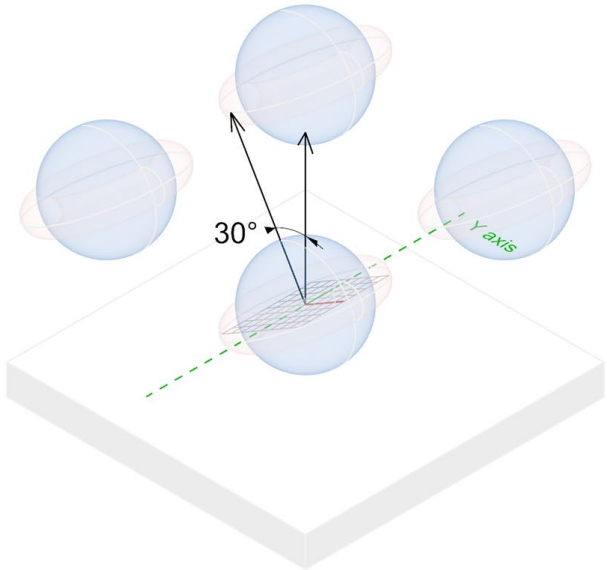
1. Constructing **spheres** and **toruses.**
2. **Rotating** a plane based on an **axis** and an **angle**.



1. Spheres and toruses

Learning objectives

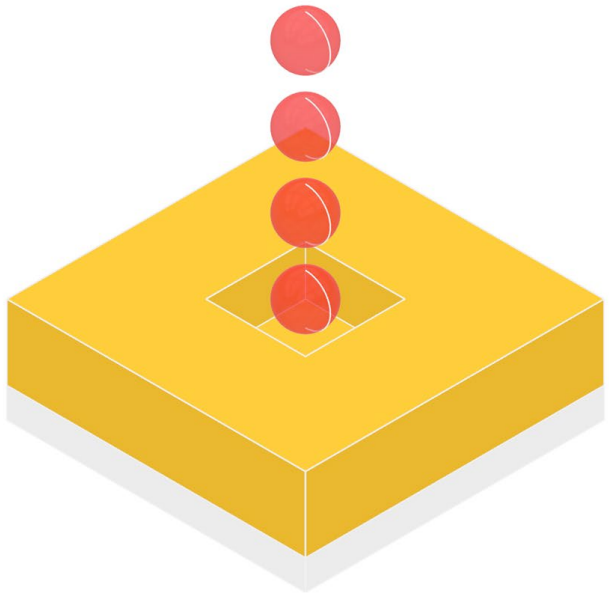
1. Constructing **spheres** and **toruses.**
2. **Rotating** a plane based on an **axis** and an **angle**.



2. Box and spheres

Learning objectives

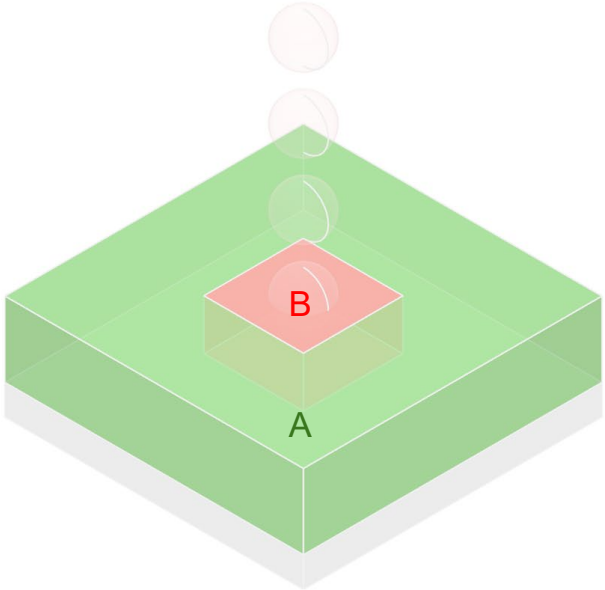
1. Boolean difference.
2. **Translation** based on vectors.



2. Box and spheres

Learning objectives

1. Boolean difference.
2. **Translation** based on vectors.

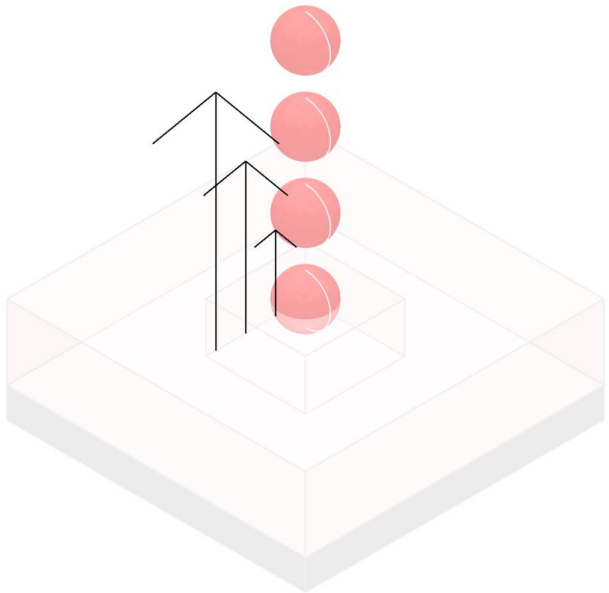


Box A - Box B

2. Box and spheres

Learning objectives

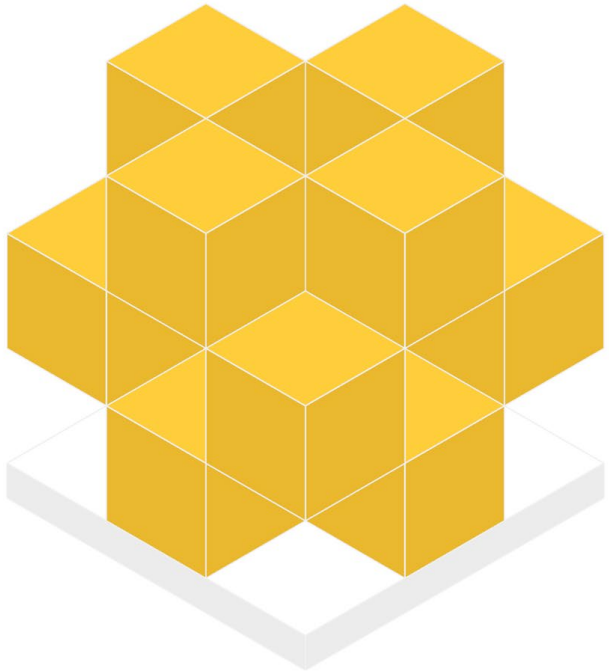
1. Boolean difference.
2. **Translation** based on vectors.



3. Chequered boxes

Learning objectives

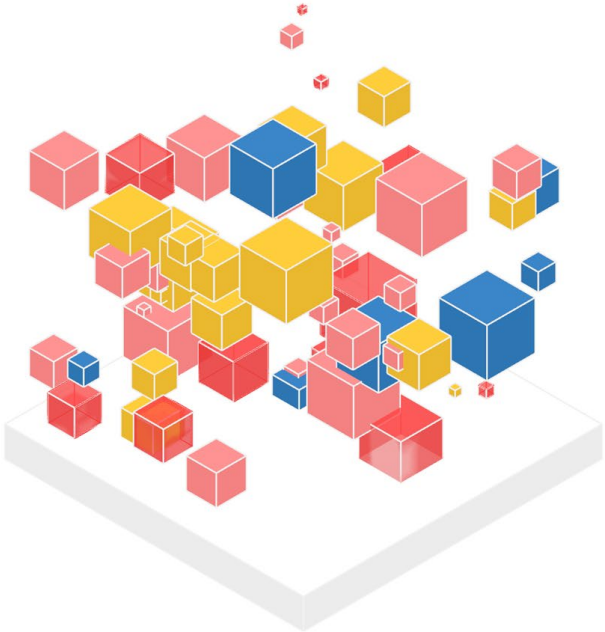
1. Extending Chequered squares in 2D to 3D.
2. Recap: Using if conditions within for loops



4. Point cloud

Learning objectives

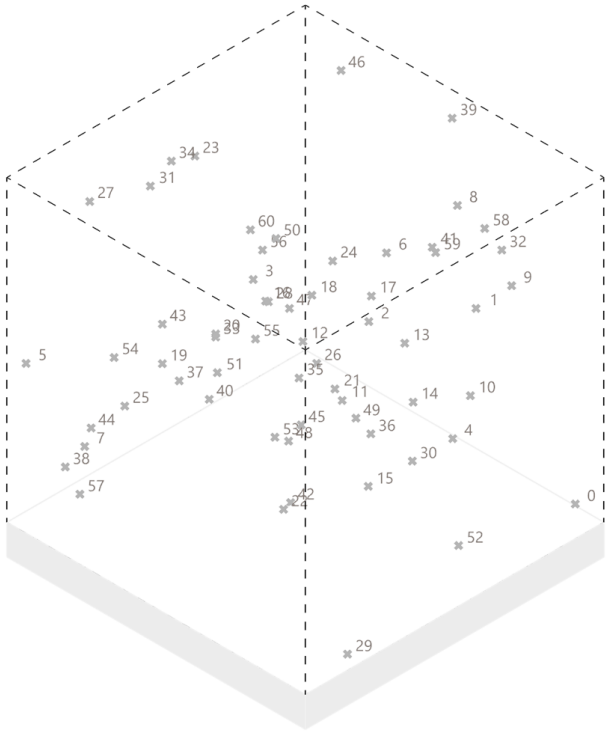
1. Creating random points within a domain.
2. Understanding scale transformation.
3. Creating random colors using script.



4. Point cloud

Learning objectives

1. Creating random points within a domain.
2. Understanding scale transformation.
3. Creating random colors using script.

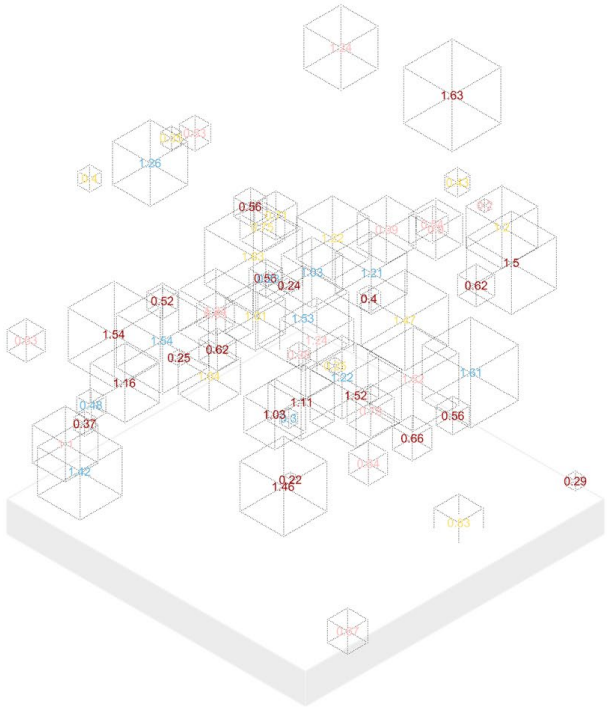


Points

4. Point cloud

Learning objectives

1. Creating random points within a domain.
2. Understanding scale transformation.
3. Creating random colors using script.

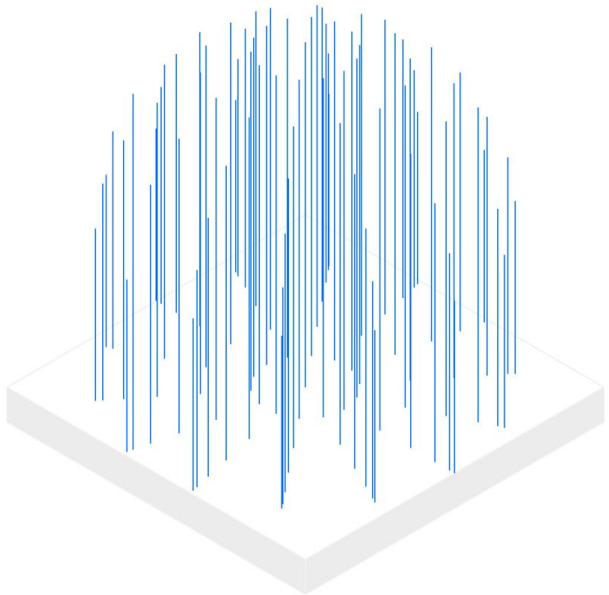


Scales and colors

5. Point traces

Learning objectives

1. **Splitting a sphere** to get the hemisphere
2. Getting **points on a surface** using **u, v** value pairs
3. **Projecting points** on a brep



5. Point traces

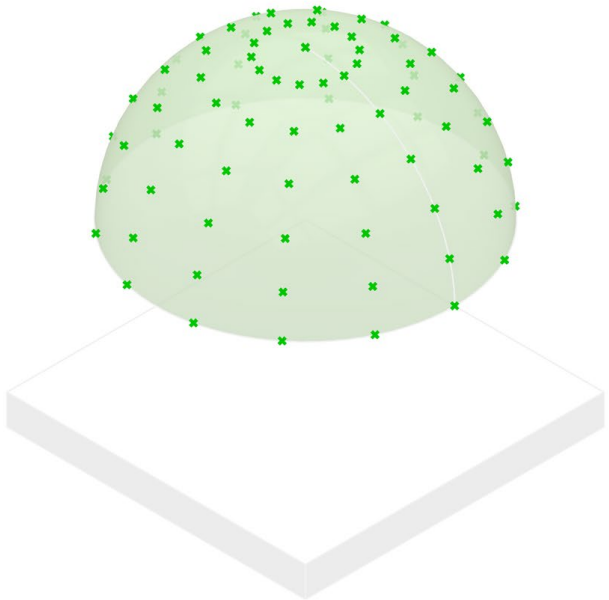
Learning objectives

1. **Splitting a sphere** to get the hemisphere
2. Getting **points on a surface** using **u, v** value pairs
3. **Projecting points** on a brep

5. Point traces

Learning objectives

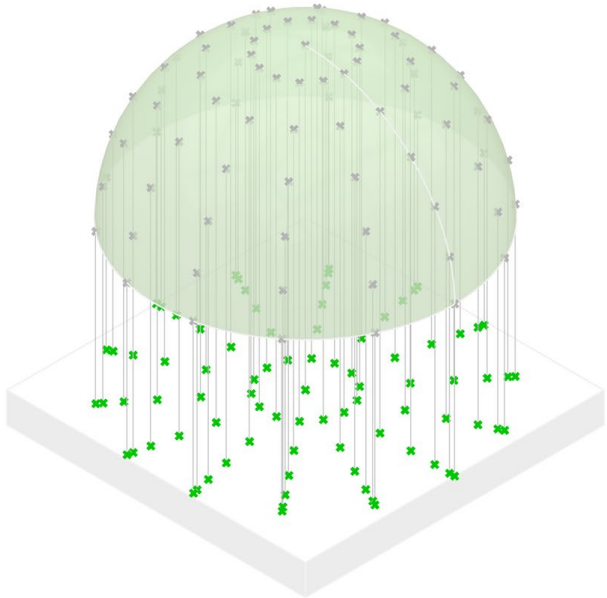
1. **Splitting a sphere** to get the hemisphere
2. Getting **points on a surface** using **u, v** value pairs
3. **Projecting points** on a brep



5. Point traces

Learning objectives

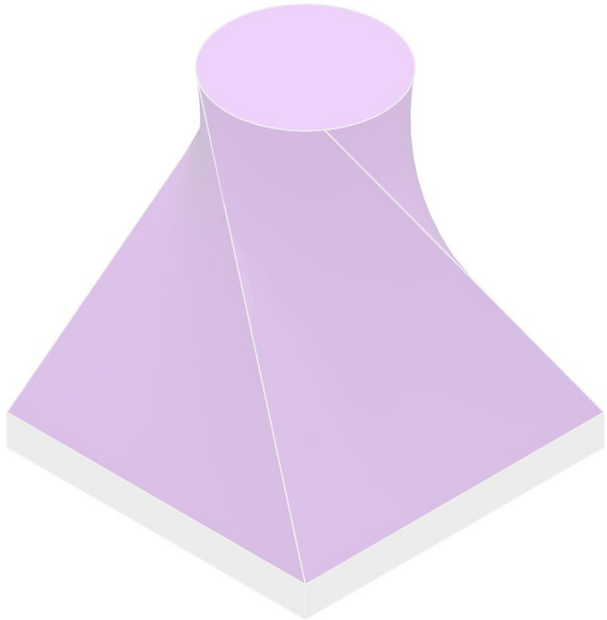
1. **Splitting a sphere** to get the hemisphere
2. Getting **points on a surface** using **u, v** value pairs
3. **Projecting points** on a brep



6. Blend surface

Learning objectives

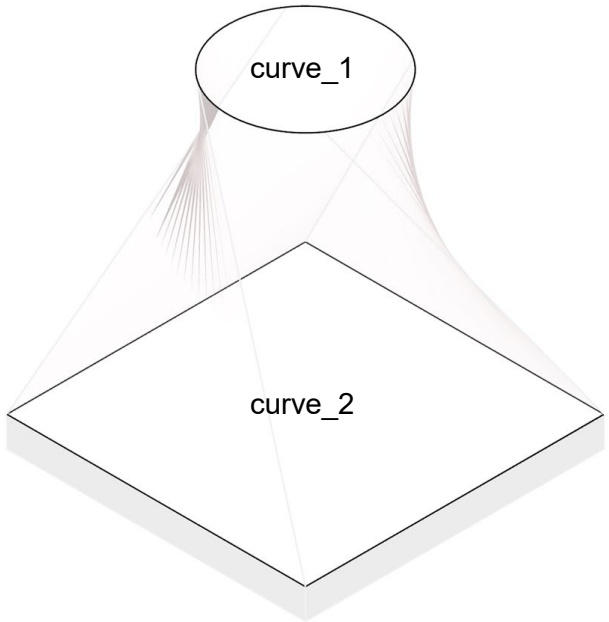
Creating a solid by blending
2 curves



6. Blend surface

Learning objectives

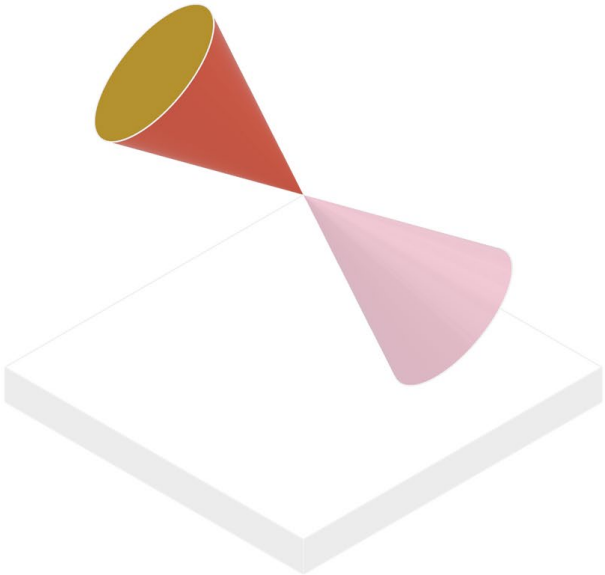
Creating a solid by blending
2 curves



7. Mirrored cones

Learning objectives

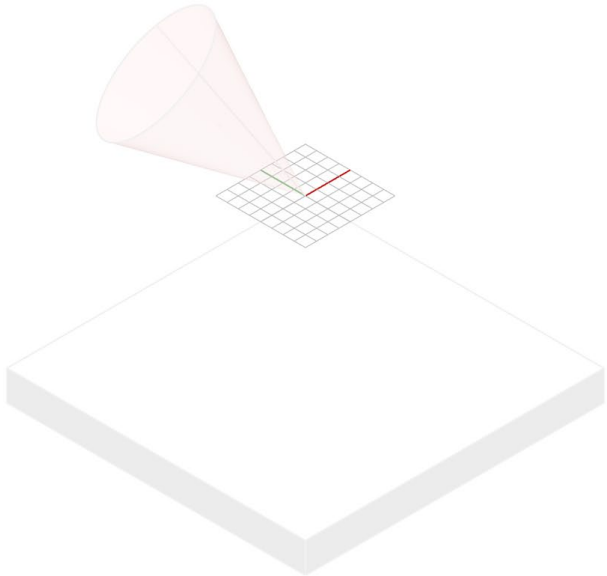
1. Constructing **cones**.
2. Using **Mirror transformation.**



7. Mirrored cones

Learning objectives

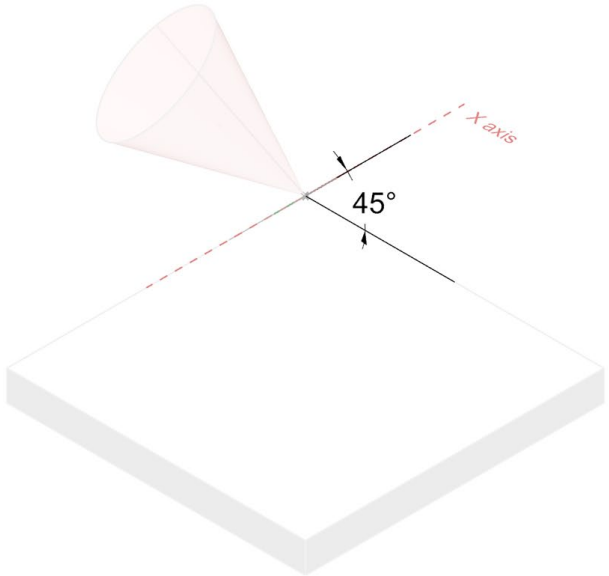
1. Constructing **cones**.
2. Using **Mirror transformation.**



7. Mirrored cones

Learning objectives

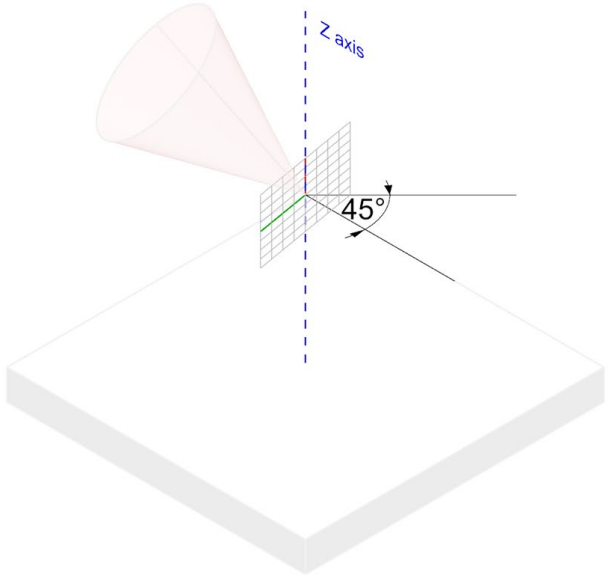
1. Constructing **cones**.
2. Using **Mirror transformation.**



7. Mirrored cones

Learning objectives

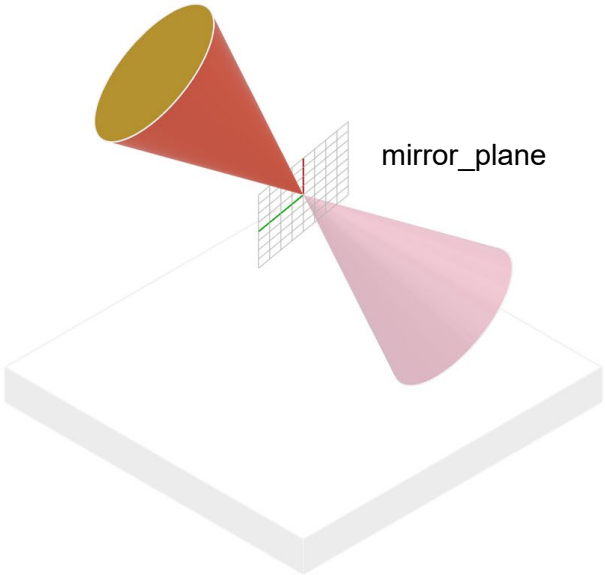
1. Constructing **cones**.
2. Using **Mirror transformation.**



7. Mirrored cones

Learning objectives

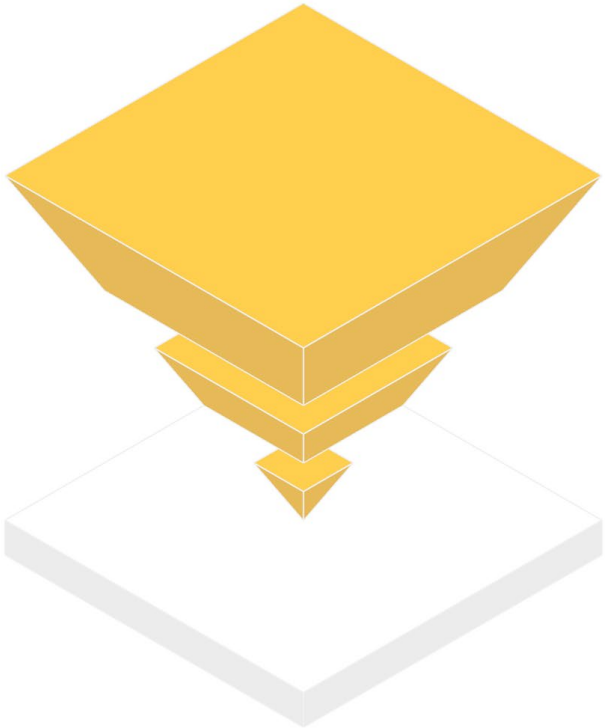
1. Constructing **cones**.
2. Using **Mirror transformation.**



8. Pyramid parts

Learning objectives

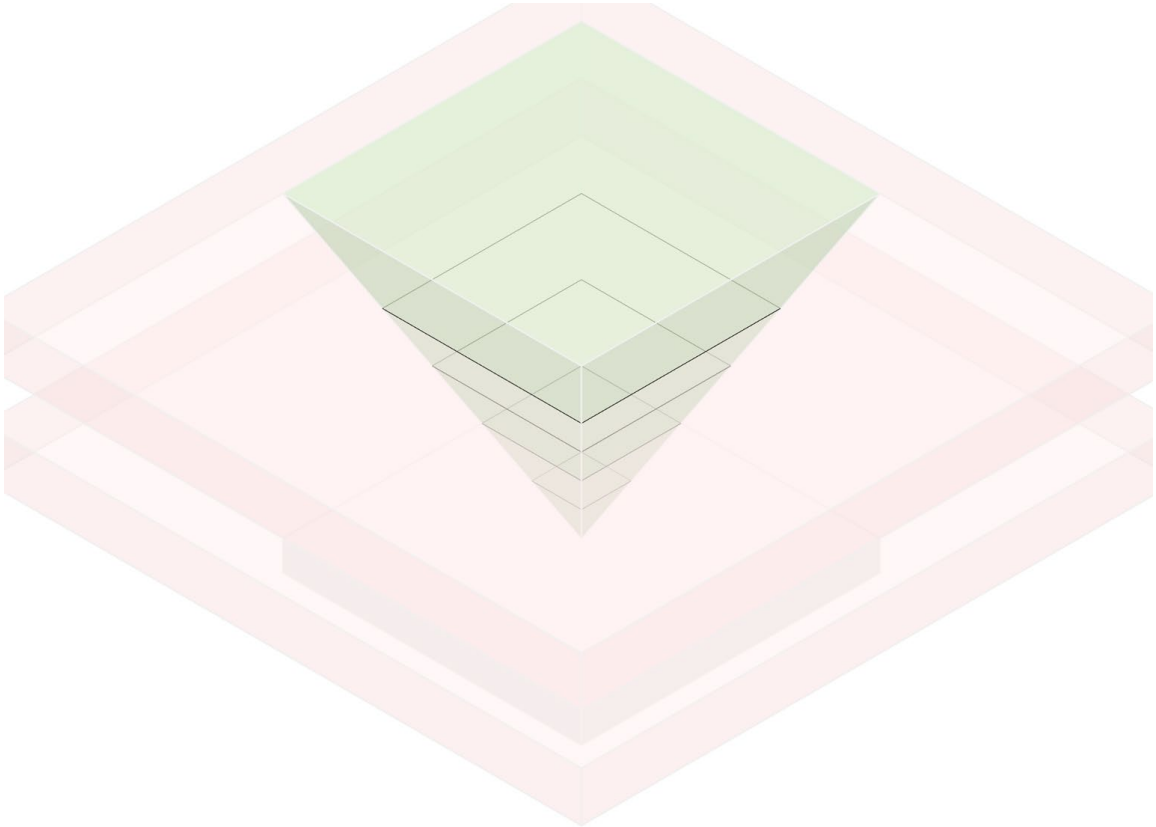
Recap: Boolean difference



8. Pyramid parts

Learning objectives

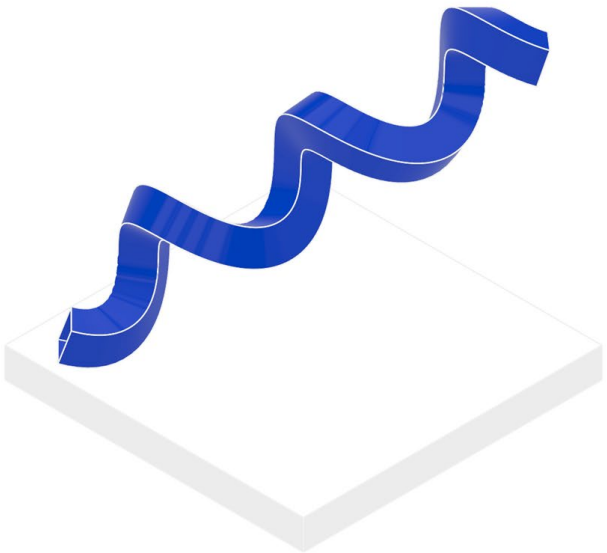
Recap: Boolean difference



9. Sweep

Learning objectives

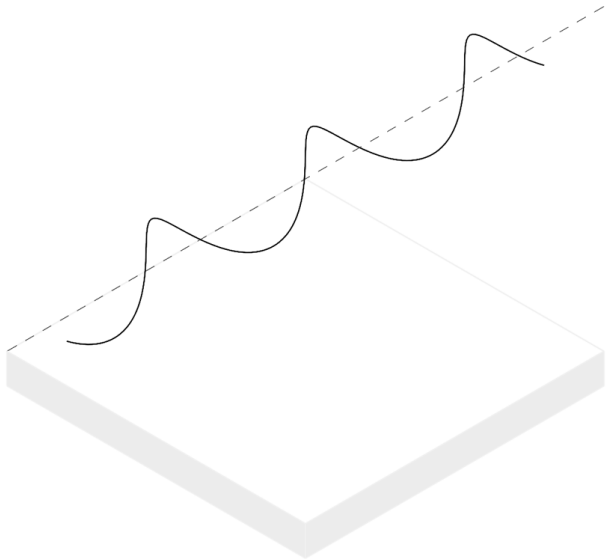
1. Construct a spiral
2. Construct a sweep



9. Sweep

Learning objectives

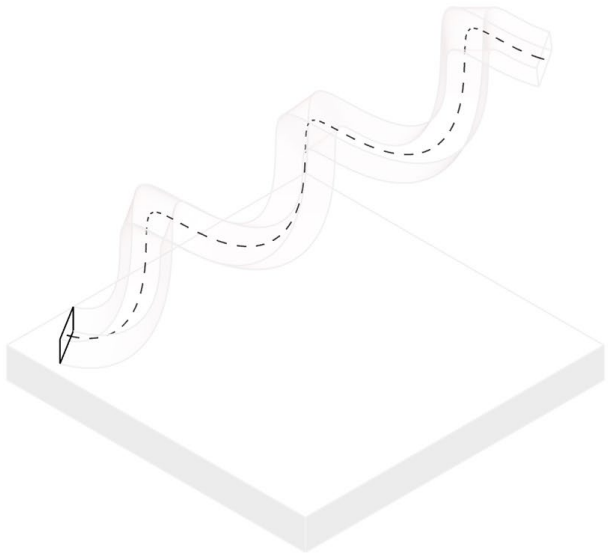
1. Construct a spiral
2. Construct a sweep



9. Sweep

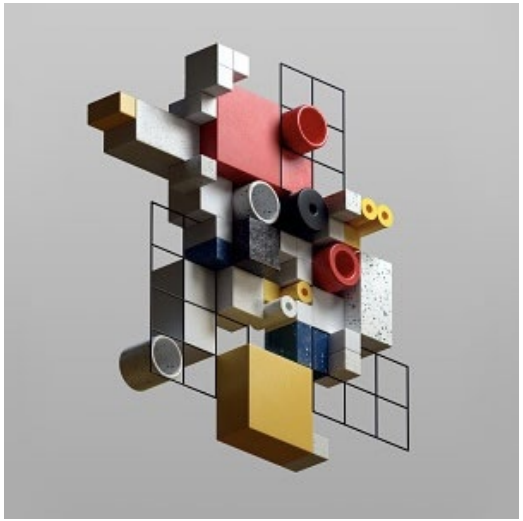
Learning objectives

1. Construct a spiral
2. Construct a sweep



Assignment 1

Bauhaus 3D Composition





Feedback session: 02.11.2026

Short presentation + Working time

Submission deadline: 09.11.2026

Online submission

Assignment 1

1 Slide Presentation

Includes:

- Images or Video of the parametric variations
- Explanation of the design *parameters*

1st Submission via Moodle (02.11)

You will present this in the feedback session! (02.11)

2nd Submission via Moodle (09.11)

Grasshopper + src code

Submission via GitHub (09.11)

Find the instructions on Moodle.

Look into a RhinoCommon API for further geometries and functions.

Enjoy creating!